Introduction to VLSI Design, 2020

Homework 2: derivations, CMOS circuits, and code

The assignment itself is analytical. Runnable verification code and HSPICE decks are included to meet the requested programming deliverable. The local process model stays at `D:/cic018.l` and is not distributed.

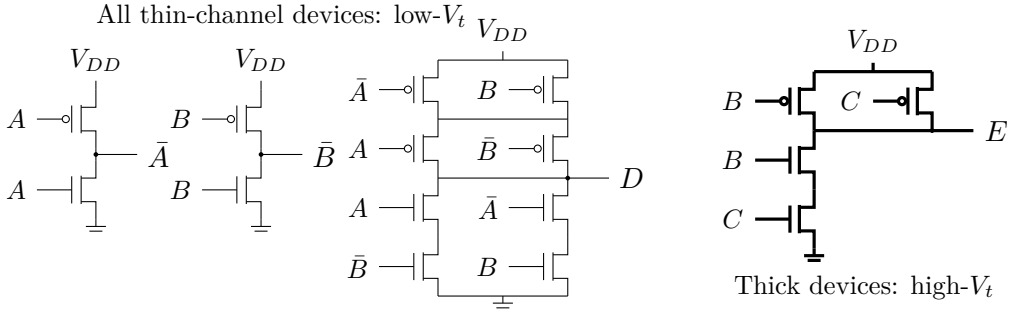
1. Multi-V_t design

The circuit in Fig. 1 implements

$$D = \bar{A} \oplus B = AB + \bar{A}\bar{B}, \qquad E = \overline{BC}.$$

In the given gate-delay abstraction, $A \to$ INV $\to$ XOR $\to D$ has delay 2; $B \to$ XOR $\to D$ and $B/C \to$ NAND $\to E$ have delay 1. Keep the D cone low-V_t. The E NAND has one unit of slack and can use high-V_t devices *provided its characterized total delay is at most 2 units*. Then the circuit's maximum delay remains 2 units while the NAND's subthreshold leakage decreases.

The problem does not give the high-V_t delay or models, so preservation of timing is conditional on that bound. A physical guarantee for an arbitrary high-V_t process would require characterized delays. Adding an inverter for $\bar{B}$ when implementing the XOR at transistor level adds a two-stage B path; keep that inverter low-V_t as well. The transistor schematic below implements D as XNOR directly using generated $\bar{A}, \bar{B}$.



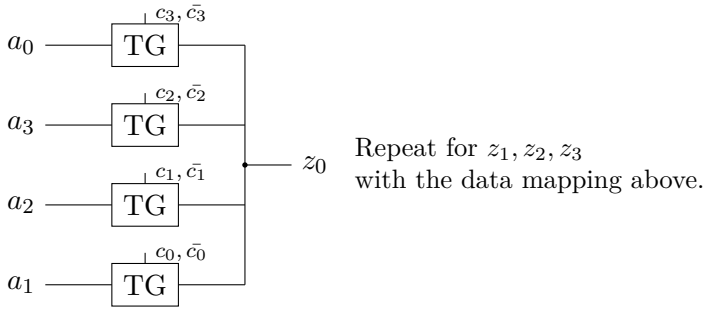
The upper figure contains 12 low-V_t devices; the E NAND contains 4 high-V_t devices. `dualvt.sp` records this topology with explicit LVT/HVT model names. CIC018 TT supplies no separate characterized high-V_t pair for this task; the file deliberately does not alias ordinary devices as high-V_t devices.

2. CMOS controlled rotator

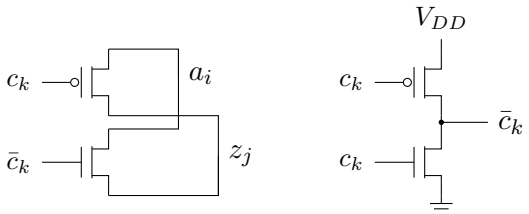
Exactly one of c_0, c_1, c_2, c_3 is 0 in the specified table. Write $s_k = \bar{c}_k$. Each output uses four parallel transmission gates, one selected by each s_k:

$$
\begin{aligned}
z_0 &= s_3a_0 + s_2a_3 + s_1a_2 + s_0a_1,\\
z_1 &= s_3a_1 + s_2a_0 + s_1a_3 + s_0a_2,\\
z_2 &= s_3a_2 + s_2a_1 + s_1a_0 + s_0a_3,\\
z_3 &= s_3a_3 + s_2a_2 + s_1a_1 + s_0a_0.
\end{aligned}
$$

Equivalently, for $c_k = 0$, $z_j = a_{(j+k+1) \bmod 4}$. This matches the original table in the order c_3, c_2, c_1, c_0.



Transistor-level TG and shared control inverter. The PMOS gate uses c_k and the NMOS gate uses $\bar{c}_k$. Both devices conduct when $c_k = 0$, so either data rail passes without a threshold-voltage loss. PMOS bodies connect to V_{DD}, NMOS bodies to ground.



Parallel n/p channel pair; body ties omitted

There are 16 TGs (32 MOSFETs) plus four shared control inverters (8 MOSFETs), for **40 transistors**. The supplied handwritten solution uses cascaded select paths; the direct one-hot arrangement is simpler and uses fewer devices. For invalid controls, multiple selected TGs can short unequal ideal data drivers or no TG can drive the output; the assignment specifies only the four valid control words. Control changes are sampled after settling.

3. Pseudo-NMOS logic gate and sizing

The PMOS is always on (gate tied to ground). The two NMOS devices are parallel. Thus the intended logic function is

$$\boxed{V_o = \overline{A + B} \quad \text{(a ratioed two-input NOR).}}$$

Let $S_p = (W/L)_p$, $S_n = (W/L)_n$, and $r = S_n/S_p$ with equal NMOS sizes. For equal original sizes $R_p = 2R_n$; scaling W/L scales on-resistance inversely. Therefore

$$\frac{R'_p}{R'_n} = 2r.$$

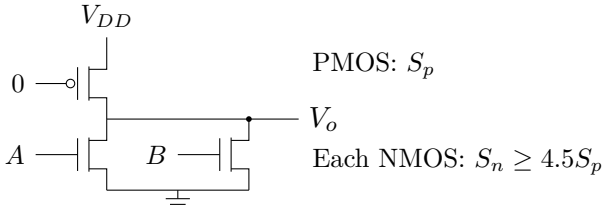
For $AB = 01$ or 10, only one NMOS conducts; this is the worst low level:

$$\frac{V_{OL}}{V_{DD}} = \frac{R'_n}{R'_p + R'_n} = \frac{1}{2r+1} \le 0.1 \quad \Longrightarrow \quad \boxed{r \ge \frac{9}{2}.}$$

For $AB = 11$, two NMOS resistances are in parallel, giving

$$\frac{V_o}{V_{DD}} = \frac{1}{4r+1}.$$

For $AB = 00$, the pull-down is off and $V_o = V_{DD}$. At $r = 4.5$, the one-on cases are $0.1V_{DD}$ and the both-on case is $V_{DD}/19 \approx 0.05263V_{DD}$. A concrete minimum design is $L_p = L_n = 0.18\,\mu$m, $W_p = 0.5\,\mu$m and $W_{nA} = W_{nB} = 2.25\,\mu$m. Choosing $W_n = 2.5\,\mu$m ($r = 5$) provides a small margin within the specified resistance model.



When the output is low, this ratioed gate consumes static current. In the ideal example $R_n = 1\,\text{k}\Omega$ for an original equal-size NMOS and $R_p = 2\,\text{k}\Omega$, the chosen $r = 4.5$ gives $I_{01} = I_{10} = 0.81$ mA and $I_{11} \approx 0.8526$ mA at $V_{DD} = 1.8$ V. Absolute resistance and current values are illustrative; the ratio bound is independent of that chosen resistance.

Reconciliation with the supplied answer

The four-page `HW2_answer.pdf`, by J.-M. Wu (107501503), correctly identifies XNOR/NAND, the rotator table, and NOR. Its first sizing derivation reverses the strength ratio and suggests $S_n/S_p = 2/9$. The final handwritten correction obtains 9/2 for one conducting NMOS and 9/4 for two. The correct *common design* is the stronger one-on constraint, $S_n/S_p \ge 9/2$. The present solution retains that corrected result, makes the timing assumption explicit, and replaces the larger rotator with a shared-control 40T design.

Reproducible numerical evidence

`python reproduce.py -model D:/cic018.l` generates the tables and decks and runs HSPICE. `-skip-hspice` runs the independent analytical checks only. The 64 valid rotator cases (16 data words for each of four controls) all passed in CIC018 TT at 1.8 V and 25 C with 1 fF on each output; highs were required to be at least 1.6 V and lows at most 0.2 V. The exported per-vector voltages are in `q2_electrical.csv`.

The ideal-resistor HSPICE results exactly matched the separate closed-form calculation within 10^{-8} V. A supplementary nonlinear TT experiment was also run. These BSIM3 voltages are measured results for its declared sizes, not replacements for the assignment's fixed-resistance formula:

r	V_{00} (V)	V_{01} (V)	V_{10} (V)	V_{11} (V)
1.0	1.800000	0.176482	0.176482	0.078389
4.5	1.800000	0.032241	0.032241	0.015732
5.0	1.800000	0.028826	0.028826	0.014100

The measured supply currents are also saved in `q3_cic018_tt_measurements.csv`. No electrical timing claim is made for Q1's unspecified high-V_t models. `q1_truth.csv` verifies all 8 input combinations and `q1_timing.csv` states the gate-delay allocation and bound.